BioMTL Dataset Documentation

Release v1.0

Dataset name: BioMTL
Contact for data requests: biomtl@163.com
Code: https://github.com/PPGPT/AAAI2026

BioMTL is released for non-commercial academic research under the accompanying Data Use Agreement. This document describes the dataset contents, structure, labels, and known limitations.

1. Overview

BioMTL is a PPG-centric benchmark for multi-task representation learning and physiological monitoring. The dataset contains 1,808 four-second fingertip-PPG windows from 172 subjects. Each window inherits subject-level labels for four continuous tasks and three binary tasks.

Item	Value
Subjects	172 (age 21-96 years)
PPG windows	1,808
Signal	Fingertip PPG, 1000 Hz, 4 s windows (4000 samples)
Tasks	SBP, DBP, BG, HR, BMI, stress, hypertension, type-2 diabetes
Collection span	285 days
Release format	NumPy arrays and CSV metadata

2. Archive structure

File	Description
data/signals.npy	PPG signal array, dtype float32, shape (1808, 4000).
data/window_subject.npy	Subject id for each signal row, dtype int32, shape (1808,).
data/labels.csv	One row per subject, including labels for all seven tasks and demographic variables.
data/example_window.csv	Rounded CSV copy of the first PPG window for inspection and plotting.
documents/en/	English PDF documentation and agreement.
documents/zh/	Chinese PDF documentation and agreement.

Alignment

Row *i* of signals.npy corresponds to element *i* of window_subject.npy. The subject id is used to join each window to labels.csv. Labels are provided at the subject level; a subject's windows share the same label values.

3. Signal format and preprocessing

- Shape: (1808, 4000), dtype float32.

- Sampling rate: 1000 Hz; each row is a 4 s window containing 4000 samples.

- Released signals are band-pass filtered AC-amplitude waveforms. The acquisition workflow used a zero-phase, forward-backward fourth-order Butterworth filter with a 0.5-8 Hz passband.

- Per-window z-score normalization should be applied at model input.

- Each subject contributes 10 or 11 windows (84 subjects with 10 windows and 88 subjects with 11 windows).

- example_window.csv is a rounded human-readable copy of the first signal row and should not be used as a replacement for signals.npy.

4. Label dictionary

Column	Type	Unit / coding	Task group
subject_id	int	identifier	key
n_windows	int	count	number of windows for the subject
bg	float	mmol/L	blood glucose regression
sbp, dbp	int	mmHg	blood pressure regression
hr	int	bpm	heart rate regression
bmi	float	kg/m^2	BMI regression
weight	float	kg	demographic
height	float	m	demographic
age	int	years	demographic
sex	int	0 = female, 1 = male	demographic
stress	int	0/1	mental stress classification
htn	int	0/1	hypertension classification
t2dm	int	0/1	type-2 diabetes classification

Sex counts are 100 male and 72 female. Binary-label positive counts are: stress 50, hypertension 71, and type-2 diabetes 40.

5. Cohort statistics

Metric	Mean	Min	Max	SD
Weight (kg)	65.80	45	97	12.35
Height (m)	1.71	1.55	1.87	0.09
Age (years)	48.97	21	96	25.41
BMI (kg/m^2)	22.41	16.84	36.73	3.41
HR (bpm)	80.01	55	116	13.09
SBP (mmHg)	119.51	82	164	18.17
DBP (mmHg)	72.35	34	105	10.95
BG (mmol/L)	6.41	4.2	16.3	2.17

6. Acquisition protocol

Data were collected under an open-environment protocol designed to preserve natural physiological and signal variation. Each PPG sample was synchronized with the associated task annotations under standardized acquisition and quality-assessment procedures.

Measure	Instrumentation
Blood glucose	Accu-Chek Performa capillary glucose meter (Roche), with daily L1/L2 quality control.
Blood pressure	Omron J751 upper-arm monitor.
PPG	EVAL-ADPD4100Z-PPG board with EVAL-ADPDUCZ MCU (Analog Devices), fingertip acquisition at 1 kHz.

7. Ethics and privacy

The study was reviewed and approved by the institutional ethics committee of the participating clinical site. Participants provided informed consent before data collection. The released dataset is de-identified and uses integer subject identifiers only. Re-identification attempts are prohibited by the Data Use Agreement.

8. Intended use

BioMTL is intended for non-commercial academic research on PPG representation learning, multi-task learning, transfer learning, few-sample learning, and physiological-monitoring algorithms. The dataset is not intended for direct clinical diagnosis, treatment, or medical decision-making.

9. Known limitations

- Labels are provided at the subject level; windows from the same subject share label values.

- The cohort is single-site and may not represent other populations, devices, or clinical settings.

- No timestamps are released.

- Released waveforms are filtered AC-amplitude signals; per-window z-score normalization should be applied at model input.

- The dataset is for algorithmic research and benchmark evaluation only.

10. Citation and use conditions

Use is governed by the Data Use Agreement. Publications or public outputs using BioMTL should cite the associated publication:

> Zhang, Z., Lu, H., & Zhao, Q. (2026). *PPGPT: Transferring Next-Token Modeling from Language to PPG Signals.* Proceedings of the AAAI Conference on Artificial Intelligence, 40(34), 28573-28581. https://doi.org/10.1609/aaai.v40i34.40088


```
@inproceedings{zhang2026ppgpt,
  title     = {PPGPT: Transferring Next-Token Modeling from Language to PPG Signals},
  author    = {Zhang, Zexing and Lu, Huimin and Zhao, Qingxin},
  booktitle = {Proceedings of the AAAI Conference on Artificial Intelligence},
  volume    = {40},
  number    = {34},
  pages     = {28573--28581},
  year      = {2026},
  doi       = {10.1609/aaai.v40i34.40088},
  url       = {https://ojs.aaai.org/index.php/AAAI/article/view/40088}
}
```